

GAMES OF LEARNING

Name of the qualification:	<i>Diploma in Information Technology in Network Management.</i>
Module Name:	<i>Work Integrated Learning (XINM 5219)</i>
Name of the company:	<i>Games of Learning (GFL)</i>
Group members:	<i>Thato Manyama ST10450306 Siboniso Maruge ST10459440 Andile Rakhombe ST10524430 Winston Dube ST10181381 Sandra Langa st10442678</i>
Slogan:	<i>Gaming minds. Bright Futures.</i>

Contents

1. Introduction	3
2. Requirements Analysis	4
User Requirements.....	4
Device Requirements	4

Application Requirements	4
Connectivity Requirements	4
Constraints and Considerations	5
4. IP Addressing Scheme.....	6
Default Gateway	6
Management Approach	7
5. Hardware & Equipment List	8
6. Wireless Network Design	12
6.1 Access Point Placement.....	12
6.2 SSID Configuration	12
6.3 Security Features	12
6.4 Performance Considerations	13
6.5 Future Scalability	13
7. Connectivity Plan (WAN/ISP)	14
7.1 Internet Service Provider	14
7.2 Primary WAN Setup	14
7.3 Site-to-Site VPN	14
7.4 Redundancy and Failover	14
7.5 Bandwidth Management	15
7.6 Future Considerations	15
8. Security Measures	15
8.1 Network Edge & Perimeter	15
8.2 Internal Segmentation (VLANs & ACLs)	13
8.3 Wi-Fi Security	16
8.4 Identity, Access & Endpoint	16
8.5 Secure Services & Data	17
8.6 Patch & Vulnerability Management	17

8.7 Monitoring, Logging & Alerting	17
8.8 Backup & Disaster Recovery	17
8.9 Physical & Environmental Security	17
8.10 Change & Incident Management	18
8.11 Printer & NVR Hardening	18
10. Scalability & Future Growth.....	19
10.1 User Growth	20
10.2 Network Expansion	20
10.3 Server & Storage Growth	20
10.4 Internet & WAN Upgrades	20
10.5 Security & Compliance	20
11. Testing & Validation Plan	21
11.1 Connectivity Testing	21
11.2 DHCP & IP Addressing Validation	21
11.3 Wireless Network Testing	21
11.4 Security Testing.....	22
11.5 Performance & Load Testing	22
11.6 Backup & Recovery Validation	22
11.7 Disaster Recovery Testing	22
11.8 Documentation & Handover	22

1. Introduction

The purpose of this Network Design Document (NDD) is to outline the technical and infrastructural requirements for the establishment of a secure, reliable, and scalable network at Site B of Games for Learning (GFL). This document provides a detailed plan

that covers the topology, IP addressing, hardware, wireless connectivity, security, and disaster recovery measures required for successful implementation.

The design is intended to support both the administrative and development functions of EN, ensuring that staff have access to the necessary resources for daily operations while maintaining a strong focus on network performance and security.

The scope of this document is limited to the deployment of the internal Local Area Network (LAN) and associated infrastructure at Site B, while ensuring seamless integration with Site A via a secure Virtual Private Network (VPN). The design also accounts for scalability to accommodate future growth in staff numbers, additional servers, and the expansion of services as the company grows.

The key objectives of this design are as follows:

- Ensure high availability and reliability of network resources.
- Provide secure and segregated access to network services.
- Support the performance demands of a gaming development environment.
- Facilitate disaster recovery and business continuity.
- Enable scalability for future expansion.

2. Requirements Analysis

The network design for Site B is based on the operational needs of 17 users, consisting of both developers and administrative staff. The analysis of requirements is broken down into the following areas:

User Requirements

- 17 staff members to be supported at Site B.
- Developers require high-performance workstations capable of running gaming development and design software.
- Administrative staff require standard workstations for productivity, communication, and management tasks.

Device Requirements

- 17 workstations (combination of developer and admin PCs).
- 3 servers (File server, Database server, NAS backup).
- 3 printers (networked with static IP addresses).

- 3 Wi-Fi access points to provide wireless connectivity across the office space.
- 6 CCTV IP cameras for physical security.
- 1 NVR (Network Video Recorder) server for CCTV recording and management.
- 1 smart TV for meetings, training, and presentations.

Application Requirements

- Development tools (game engines, code editors, graphic design software).
- Business productivity applications (email, office suites, accounting software).
- Collaboration tools (internal chat, video conferencing).
- Remote access to servers and databases via VPN.

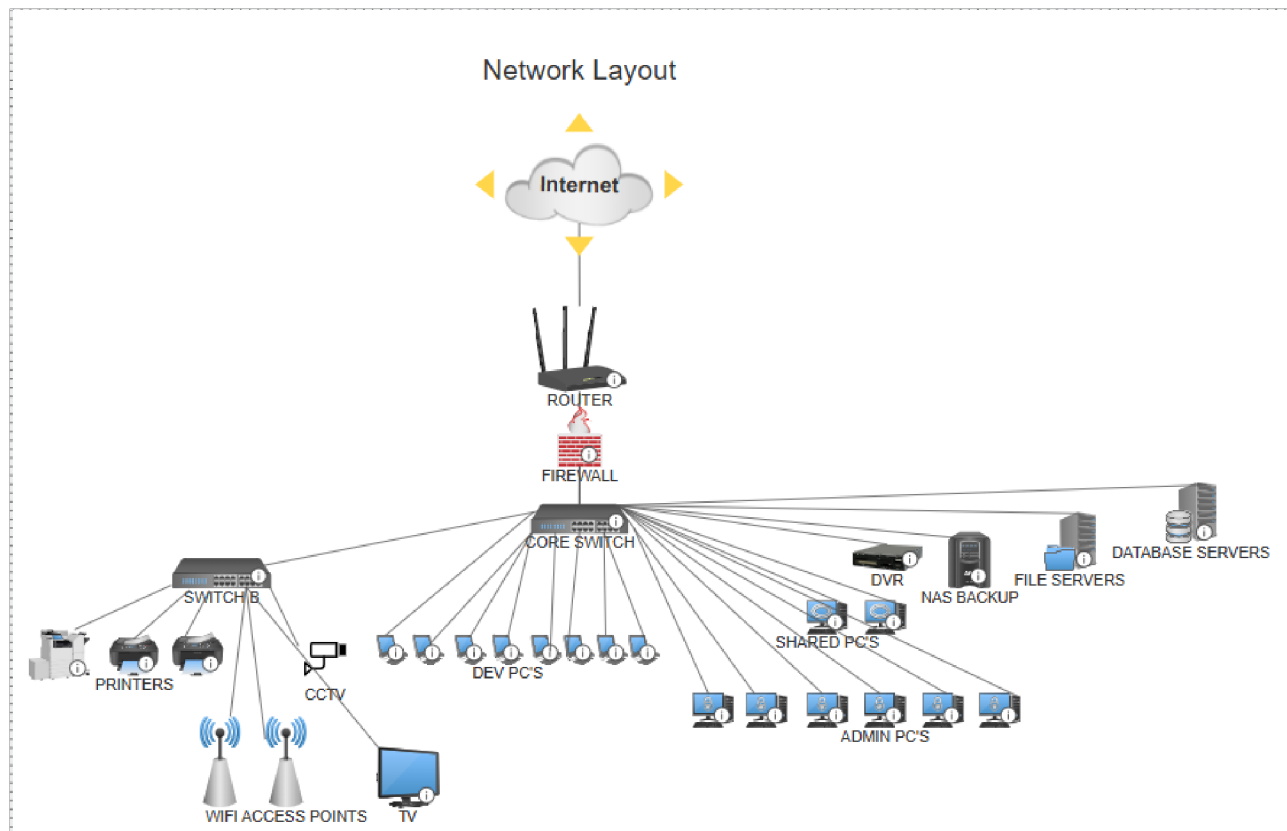
Connectivity Requirements

- Dedicated fibre internet connection provided by Telkom.
- Minimum of 100 Mbps symmetrical bandwidth for upload/download.
- Secure VPN tunnel to connect Site B with Site A.
- Full Wi-Fi coverage across the building with separate SSIDs for staff and guest users.

Constraints and Considerations

- Budget limitations require cost-effective but reliable equipment choices.
- Security is critical due to the sensitivity of game development projects.
- Scalability must be ensured to allow for future growth, including additional staff, servers, and services.

3. NETWORK TOPOLOGY DESIGN



4. IP Addressing Scheme

The Site B network will implement a structured IP addressing plan to ensure efficient allocation of resources, ease of management, and prevention of address conflicts. A Class C private IP range has been chosen, with subnetting applied to separate devices logically.

The chosen subnet is:

- Network ID: 192.168.20.0/24
- Subnet Mask: 255.255.255.0
- Total Usable Addresses: 254

A combination of static and dynamic IP addressing will be used. Static addresses will be allocated to critical infrastructure devices such as servers, printers, and access points. Dynamic Host Configuration Protocol (DHCP) will be used for user devices and mobile connections to streamline management.

Category	IP Range	Notes
Default Gateway	192.168.20.1	Router/firewall gateway
Servers	192.168.20.10 – .19	File Server, Database Server, NAS, NVR
Printers	192.168.20.50 – .55	Static IPs for 3 network printers
Admin PCs	192.168.20.60 – .69	Static/DHCP reservations for 6 admin PCs
Developer PCs	192.168.20.70 – .79	Static/DHCP reservations for 9 developer PCs
Wi-Fi APs	192.168.20.80 – .85	Static IPs for 3 access points
CCTV Cameras	192.168.20.90 – .99	6 IP cameras on VLAN 40
Dynamic Pool	192.168.20.100 – .199	For mobile devices, laptops, and guest Wi-Fi
Reserved	192.168.20.200 – .254	Expansion, testing, and future devices

Default Gateway

All devices will use 192.168.20.1 as their default gateway, which is assigned to the router/firewall.

Management Approach

- DHCP will be configured to manage dynamic allocations within the defined pool.
- Static reservations will be used for workstations and critical infrastructure.
- A central IP address register will be maintained to document assignments and avoid conflicts.

This IP addressing scheme ensures that every device is uniquely identifiable, supports secure VLAN segmentation, and allows for straightforward scalability as Site B expands.

5. Hardware & Equipment List

The following hardware has been selected to meet the operational and security requirements of Site B. Equipment is chosen with a balance of performance, reliability, and scalability in mind, while remaining within budgetary constraints.

End-User Devices

Item	Qty	Unit Price (ZAR)	Total (ZAR)	Notes
Developer PCs	9	R15,000	R135,000	High-performance (gaming capable)
Admin PCs	6	R10,000	R60,000	Standard productivity machines
Shared/Hot desk PCs	2	R10,000	R20,000	Flexible/intern use
Subtotal	17		R215,000	

Servers & Storage

Item	Qty	Unit Price (ZAR)	Total (ZAR)	Notes
File Server	1	R40,000	R40,000	Central file storage
Database Server	1	R50,000	R50,000	For dev databases
NAS Backup Device	1	R30,000	R30,000	Redundant storage & recovery
NVR Server (CCTV)	1	R25,000	R25,000	CCTV footage storage
Subtotal	4		R145,000	

Networking Equipment

item	Qty	Unit Price (ZAR)	Total (ZAR)	Notes
Router (Enterprise)	1	R20,000	R20,000	With VPN support
Firewall Appliance	1	R25,000	R25,000	Security layer
Core Switch (48port)	1	R30,000	R30,000	Layer 3, VLAN support

Access Switch (24port)	1	R20,000	R20,000	For APs, printers, CCTV
Wi-Fi Access Points	3	R5,000	R15,000	WPA3, dual-band
Patch Panels & Cabling	1 lot	R15,000	R15,000	CAT6 cabling, terminations
Subtotal			R125,000	

Peripherals & Office Devices

Item	Qty	Unit Price (ZAR)	Total (ZAR)	Notes
Printers (MFP)	3	R7,000	R21,000	Networked multifunction printers
Smart TV	1	R7,000	R7,000	Meeting room display
CCTV Cameras	6	R3,000	R18,000	IP PoE cameras
UPS Units	2	R10,000	R20,000	Backup power for servers
Subtotal			R66,000	

Furniture & Safety

Item	Qty	Unit Price (ZAR)	Total (ZAR)	Notes
Desks	20	R2,500	R50,000	For PCs and staff seating
Chairs (Ergonomic)	20	R2,000	R40,000	Staff seating
Storage Cabinets	5	R3,000	R15,000	Document & equipment storage
Fire Extinguishers	5	R1,500	R7,500	Placed strategically
Smoke Detectors	10	R600	R6,000	Safety compliance
Emergency Lights	5	R1,000	R5,000	Evacuation safety
Subtotal			R123,500	

Cost Summary

Category	Total (ZAR)
End-User Devices	R215,000
Servers & Storage	R145,000
Networking Equipment	R125,000
Peripherals & Office Devices	R66,000
Furniture & Safety	R123,500
Grand Total	R674,500

6. Wireless Network Design

The wireless network at Site B is designed to provide high-speed, secure, and reliable connectivity for staff, mobile devices, and guests (Cisco, 2020; Odom, 2021). The design

ensures full coverage across the premises and is segmented using VLANs to maintain performance and security (Stallings, 2021).

6.1 Access Point Placement

Three enterprise-grade Wi-Fi Access Points (APs) will be deployed. Placement ensures maximum coverage with minimal overlap (Cisco, 2020):

- AP1 – Reception and administrative area.
 - AP2 – Developer workspace.
 - AP3 – Meeting/conference room and common area.
- APs will be connected via PoE-enabled switches, eliminating the need for external power supplies (Odom, 2021).

6.2 SSID Configuration

Two distinct SSIDs will be configured to separate internal and external users:

1. GFL-Staff: Used by developers and administrative staff, encrypted with WPA3Enterprise, segmented into VLAN 20 (Developers) and VLAN 10 (Admins) with DHCP reservations for staff devices (Stallings, 2021).
2. GFL-Guest: For visitors and temporary users, encrypted with WPA2/WPA3Personal, assigned to VLAN 30 (Guest Wi-Fi), and configured with Internet-only access with bandwidth throttling to prevent network congestion (Cisco, 2020).

6.3 Security Features

MAC address filtering, 802.1X authentication using RADIUS, guest network isolation, and regular firmware updates are implemented to ensure security (Odom, 2021; Stallings, 2021).

6.4 Performance Considerations

Each AP supports dual-band 2.4 GHz and 5 GHz frequencies. Load balancing and roaming are enabled to prevent bottlenecks and ensure seamless mobility for staff. APs support Wi-Fi 6 for high throughput and low latency (Cisco, 2020).

6.5 Future Scalability

The design allows additional APs and SSIDs for new departments or IoT devices, with VLAN segmentation maintaining network performance and security (Stallings, 2021).

7. Connectivity Plan (WAN/ISP)

Reliable connectivity is critical for Site B, including secure communication with Site A (Odom, 2021).

7.1 Internet Service Provider

Telkom Fibre Business is the selected ISP, with a provisioned 100 Mbps symmetrical connection terminated at Site B's server room. A static public IP supports VPN and remote access (Cisco, 2020).

7.2 Primary WAN Setup

ISP fibre terminates into an enterprise router. Traffic passes through a dedicated firewall, with QoS prioritizing gaming development, video conferencing, and VoIP over guest traffic (Stallings, 2021).

7.3 Site-to-Site VPN

A secure IPsec VPN tunnel links Site B to Site A, providing encrypted communication, centralized access, and remote backup replication, with AES-256 encryption (Odom, 2021).

7.4 Redundancy and Failover

A Telkom LTE failover line ensures continuity, with automatic failover handled by the firewall. Critical services remain operational during outages (Cisco, 2020).

7.5 Bandwidth Management

VLANs are prioritized based on function, with guest VLAN rate-limited to prevent congestion (Stallings, 2021).

7.6 Future Considerations

Bandwidth can be scaled, additional ISPs introduced, and cloud integration supported through VPN architecture (Odom, 2021).

8. Security Measures

Site B follows a defense-in-depth approach across network edge, internal segmentation, endpoints, identities, and the physical environment (Stallings, 2021).

8.1 Network Edge & Perimeter

Firewall appliances, NAT and port policies, threat feeds, and selective TLS inspection ensure secure operations (Cisco, 2020). Baseline firewall rules are defined for VPN, LAN/WAN, CCTV, printer, and guest traffic management (Odom, 2021).

8.2 Internal Segmentation (VLANs & ACLs)

VLAN	Purpose	Example Subnet	Key Policy
10	Admin	192.168.20.0/27	Full access to servers; blocked from Dev code repos unless needed.
20	Developers	192.168.20.32/27	Access to file/DB/NAS; no access to Guest; restricted to Printers.
30	Guest Wi-Fi	192.168.20.64/26	Internet only; block RFC1918; bandwidth limited.
40	CCTV (PoE cams)	192.168.20.128/27	Cameras → NVR only; no Internet; NVR → time/NTP + admin from VLAN10/20.
50	Infrastructure	192.168.20.160/28	APs, switches, controllers; mgmt. from Admin only.

8.3 Wi-Fi Security

- Staff SSID (GFL-Staff): WPA3-Enterprise + 802.1X/RADIUS, per-user creds, VLAN tag (10/20) (Cisco, 2020; Odom, 2021).
- Guest SSID (GFL-Guest): WPA2/3-Personal (or captive portal), VLAN 30, client isolation, rate limits (Stallings, 2021).
- RF Hygiene: Disable WPS, schedule nightly AP auto-channel, restrict 2.4 GHz if density allows (Cisco, 2020).

8.4 Identity, Access & Endpoint

- Identity: Microsoft 365/AAD + MFA for admins and remote access. Role-Based Access Control (RBAC) (Odom, 2021).
- Endpoints: Central AV/EDR, disk encryption (BitLocker), USB control (allow-list), screens lock at 10 min (Stallings, 2021).
- Servers: CIS-style hardening, least privilege, service accounts with rotated secrets (Stallings, 2021).
- Privileged Access: Admin tasks via separate admin accounts; no daily use of domain admin (Cisco, 2020).

8.5 Secure Services & Data

- Source Assets & Repos: Host on secured server; developers access via SSH/HTTPS with MFA tokens (Odom, 2021).
- File Shares: NTFS permissions (least privilege); sensitive shares read-only for non-owners (Stallings, 2021).
- Database: Encrypted at rest; limited network scope; separate DB accounts per app/service (Cisco, 2020).

8.6 Patch & Vulnerability Management

- Patching: Monthly patch window (OS, firmware for firewall/switch/AP/servers) (Odom, 2021).
- Vulnerability Scans: Quarterly internal scans; track remediation SLAs (critical: 7 days) (Stallings, 2021).
- Firmware: Stage in test first (virtualization host or lab switch) (Cisco, 2020).

8.7 Monitoring, Logging & Alerting

- Central Syslog/SIEM: Send firewall, switch, AP, server, and NVR logs to a central collector (Odom, 2021).
- Alerts: VPN failures, multiple auth failures, port flaps, IPS hits, AP rogue detection (Stallings, 2021).
- Time Sync: All devices via internal NTP; firewall peers with public NTP (Cisco, 2020).

8.8 Backup & Disaster Recovery

- 3-2-1 Backup: 3 copies, 2 media, 1 off-site (NAS + encrypted cloud) (Odom, 2021).
- RPO/RTO Targets: RPO 4 hrs. (critical shares/DB), RTO 8 hrs. for core services (Stallings, 2021).
- CCTV Retention: NVR stores 30 days on local RAID; weekly config backup off-site (Cisco, 2020).
- Recovery Tests: Quarterly restore drills (file share + DB + NVR sample) (Odom, 2021).

8.9 Physical & Environmental Security

- Server Room: Access control (RFID/pin), CCTV coverage, locked rack, tamper alarms (Stallings, 2021).
- Power: UPS per server rack; graceful shutdown; surge protection (Cisco, 2020).
- Fire Safety: ABC extinguishers, smoke/heat detection, emergency lights; see Section 10 (Odom, 2021).

8.10 Change & Incident Management

- Change Control: Ticket for network changes; maintenance windows; documented roll-back (Stallings, 2021).
- Incident Response (IR): Triage → contain → eradicate → recover → post-mortem in 5 days (Cisco, 2020).
- Contacts: On-call matrix (Network, Systems, Security) with escalation steps (Odom, 2021).

8.11 Printer & NVR Hardening

- Printers: Static IPs, admin password changed, disable unsecured protocols (FTP/SMBv1), allow printing only from VLANs 10/20 (Stallings, 2021).
- NVR & Cameras: Unique strong creds, disable P2P cloud features, no Internet egress from VLAN 40, firmware kept current (Cisco, 2020).

9. Fire Safety & Escape Measures

9.1 Fire Detection Systems

- Smoke Detectors: Ten detectors strategically placed across office areas, server room, and common spaces (Stallings, 2021).
- Heat Sensors: Installed in the server room to detect abnormal rises in temperature that could indicate equipment failure or fire risk (Odom, 2021).
- Alarm System: Integrated with detectors to alert occupants immediately and trigger evacuation procedures (Cisco, 2020).

9.2 Fire Suppression Equipment

- ABC Fire Extinguishers: Five units placed in high-risk areas (server room, kitchen, office spaces). Suitable for electrical and combustible fires (Stallings, 2021).
- CO₂ Fire Extinguisher: Dedicated to the server room, as it can extinguish electrical fires without damaging equipment (Cisco, 2020).
- Automatic Shutoffs: Critical power circuits in the server room are backed by a UPS, which can automatically shut down systems in case of overheating to prevent electrical fires (Odom, 2021).

9.3 Emergency Lighting & Signage

- Emergency Lights: Installed in hallways and near exits to guide staff during power outages or fire emergencies (Stallings, 2021).
- Illuminated Exit Signs: Clearly mark evacuation routes and assembly points (Cisco, 2020).

9.4 Escape Routes & Evacuation Plan

- Two Primary Exits: Ensure rapid evacuation of all staff (Odom, 2021).
- Evacuation Drills: To be conducted quarterly, ensuring staff familiarity with escape routes (Stallings, 2021).
- Assembly Point: A designated safe area outside the building for rollcall and accountability (Cisco, 2020).
- Accessibility Consideration: Escape routes comply with accessibility standards for mobility-impaired staff (Odom, 2021).

9.5 Integration with IT Infrastructure

- Server Room Protection: Fire-resistant doors and insulation help contain potential fire spread; CCTV cameras monitor safety equipment and evacuation areas (Stallings, 2021).
- Power Backup: UPS systems prevent abrupt shutdowns, reducing the risk of sparks or electrical surges during outages (Cisco, 2020).

9.6 Training & Compliance

- Staff will undergo annual fire safety training covering extinguisher use, evacuation protocols, and assembly procedures; safety measures comply with SANS 10400 (Odom, 2021).

10. Scalability & Future Growth

10.1 User Growth

- Current design supports 17 staff members (developers and administrators).
- Spare switch ports (30% reserved capacity) are available for an additional 10–12 users without hardware upgrades.
- Additional workstations can be integrated seamlessly by extending the DHCP scope and assigning static IPs as needed (Stallings, 2021).

10.2 Network Expansion

- Hierarchical star topology allows easy integration of additional switches if the number of devices increases significantly.
- VLAN segmentation supports the creation of new departments (e.g., QA testers, designers) without affecting existing performance.
- Wi-Fi can be expanded by adding more access points, with the existing controller and VLAN framework supporting future SSIDs (Cisco, 2020).

10.3 Server & Storage Growth

- The file server and NAS devices are configured with RAID storage and can be upgraded with additional drives.
- Virtualisation is supported on the database and file servers, allowing multiple virtual servers to run on existing hardware.
- Cloud integration (Azure/AWS) is available as an option for hybrid hosting, remote development, and backups (Odom, 2021).

10.4 Internet & WAN Upgrades

- Current Telkom fibre line provides 100 Mbps symmetrical bandwidth.
- The service can be scaled to 200 Mbps or 1 Gbps as demand grows.
- Redundant ISP connections can be added for higher availability.
- VPN framework supports multi-site expansion if new offices are opened (Stallings, 2021).

10.5 Security & Compliance

- Firewall and switch licenses support scaling of VLANs and security policies without requiring new hardware.
- Centralised logging and monitoring tools can handle increased device counts.
- Compliance with evolving cybersecurity standards (ISO 27001, POPIA) can be achieved by extending current policies (Cisco, 2020).

10.6 Physical Infrastructure

- The server room has been designed with rack space capacity to host additional servers and storage appliances.
- Furniture layout allows for more desks and staff seating as the company expands.
- Fire safety measures can be scaled with additional extinguishers, detectors, and signage as office space grows (Odom, 2021).

11. Testing & Validation Plan

11.1 Connectivity Testing

- LAN Tests: Use ping and tracert to verify communication between PCs, servers, printers, and APs; confirm VLAN segmentation is working (Stallings, 2021).
- WAN Tests: Verify internet access for staff and guest networks; test site-to-site VPN connection to Site A for latency and stability (Cisco, 2020).

11.2 DHCP & IP Addressing Validation

- Verify that the DHCP server correctly assigns IP addresses in the defined range.
- Confirm that all static IP devices (servers, printers, APs, cameras) respond at their configured addresses.
- Attempt duplicate IP assignment to ensure the system prevents conflicts (Odom, 2021).

11.3 Wireless Network Testing

- Confirm SSID broadcast for both GFL-Staff and GFL-Guest.
- Test 802.1X authentication for staff and bandwidth throttling for guest network.
- Walkthrough coverage test using Wi-Fi analyser to ensure strong signal in all areas.
- Roaming validation – confirm that devices stay connected while moving between AP zones (Stallings, 2021).

11.4 Security Testing

- Firewall Rules: Attempt to access blocked services between VLANs to confirm isolation.
- Port Scanning: Use Nmap or similar to ensure only approved ports are open.
- VPN Security: Test authentication and encryption of the Site A to Site B VPN.
- CCTV VLAN: Confirm that cameras can only communicate with the NVR and not the wider LAN (Cisco, 2020).

11.5 Performance & Load Testing

- Conduct speed tests to ensure ISP delivers contracted 100 Mbps symmetrical bandwidth.
- Simulate high usage with multiple simultaneous file transfers, video calls, and WiFi users.
- Monitor CPU, RAM, and disk usage on servers during peak workloads.
- Test printer response time under concurrent print jobs (Odom, 2021).

11.6 Backup & Recovery Validation

- Perform a test restore from NAS backup to confirm data integrity.
- Validate NVR recordings are accessible and retrievable.

- Test UPS failover: simulate power outage to confirm graceful shutdown of servers (Stallings, 2021).

11.7 Disaster Recovery Testing

- Simulate failure of the fibre link and confirm automatic failover to LTE.
- Test recovery of VPN services after outage.
- Conduct a fire drill to ensure IT equipment shuts down safely and staff follow evacuation protocols (Cisco, 2020).

11.8 Documentation & Handover

- Record all test results in a validation checklist.
- Document corrective actions for any failed test.
- Final handover report will be signed off by IT management and project stakeholders (Odom, 2021).

12. REFERENCE LIST

Cisco. (2020). *Cisco wireless LAN design guide*. Cisco Systems.
<https://www.cisco.com/c/en/us/solutions/enterprise-networks/wireless.html>

Odom, W. (2021). *CCNA wireless study guide* (3rd ed.). Cisco Press.

Stallings, W. (2021). *Network security essentials: Applications and standards* (7th ed.). Pearson.